



Effects of an automated evaluation mechanism on students' writing performance and higher-order thinking in an AR-based formative-peer-assessment learning mode

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Abstract

Proficiency in written communication is an essential prerequisite for achievement in both academic settings and broader life contexts, yet many struggle with initiating writing and engaging in deep self-reflection and text revision. To address this challenge, this study integrated augmented reality (AR) technology with formative peer assessment (FPA)-based feedback methods to facilitate students' writing performance and higher-order thinking (HOT). An empirical study was conducted with 110 Chinese pupils, who were randomly assigned to three groups: a group using formative peer assessment and teacher assessment in a traditional lecture-based learning mode (TFPA); a group using formative peer assessment and teacher assessment in an AR environment (AR-TFPA); and a group using formative peer assessment and automated writing evaluation (AWE) within an AR environment (AR-AFPA). The results indicated that the TFPA approach negatively impacted students' writing performance and HOT compared to the AR-TFPA approach. While the AR-AFPA approach positively influenced writing performance compared to the AR-TFPA approach, no significant differences were observed in HOT. To better understand how FPA facilitates the writing process, we examined the mediating role of feedback types in the relationships between FPA approaches and writing performance/higher-order thinking. Mediation analysis revealed that the affective and cognitive characteristics of received feedback significantly mediated the relationships between the adopted approaches (i.e., TFPA vs. AR-TFPA; AR-AFPA vs. AR-TFPA) and improvements in writing performance and HOT. The present study aimed to provide insights into how integrating an AWE mechanism into AR-based FPA learning mode can potentially help learners' writing performance and higher-order thinking.

Keywords Augmented reality · Automated writing evaluation · Formative peer assessment · Higher-order thinking · Mediation analysis

1 Introduction

The ability to communicate effectively in writing is vital for student success, both in educational settings and future careers (Chen et al., 2024). However, developing proficiency in writing is a complex and challenging process (De Smedt et al., 2018). Many students fail to achieve the necessary level of writing skill by the end of their K-12 education (Veiga Simão et al., 2016). They often struggle to convey abstract concepts or emotions clearly in their writing (Li et al., 2023). A significant factor contributing to these challenges is the prevalence of traditional lecture-based teaching mode with teacher feedback (Chen et al., 2024). This approach often lacks authentic, experiential scenarios, making it difficult for students to connect their prior knowledge stored in long-term memory with the writing task at hand (Wang, 2017). As a result, students frequently face uncertainty about how to begin writing (Chen et al., 2025a; Wang, 2017). Furthermore, the one-to-many teacher feedback mode often fails to provide individualized suggestions or real-time feedback due to the large number of students that must be evaluated (Brkić et al., 2024; Graham, 2019). This limitation reduces students' opportunities for deep reflection on their writing, typically resulting in superficial revisions and inadequate text improvements (Liu et al., 2023; Scherer et al., 2024).

To help young learners overcome challenges in articulating ideas and structuring their work, researchers have proposed augmented reality (AR) as a solution. AR creates embodied learning contexts that facilitate the writing process by leveraging 3D synthesized objects to enhance the visualization of complex spatial relationships and abstract concepts (Allagui, 2021; Dunleavy et al., 2009). This visualization helps students activate prior knowledge stored in long-term memory, bridging new information to their writing tasks (Li et al., 2023). AR has been shown to improve key aspects of writing, including article structure (Wang, 2017), feature descriptiveness (Li et al., 2023), word choice (Allagui, 2021), and rhetorical usage (Lin et al., 2022). Furthermore, AR fosters active participation by enabling real-time interaction between learners, virtual objects, and their physical environment (Allagui, 2021). This interactive learning approach encourages students to connect new information to their existing cognitive frameworks, leading to higher-quality compositions and improved writing skills (Rietdijk et al., 2018).

On the other hand, formative peer assessment (FPA) is recognized as an effective pedagogical strategy that promotes learners' deep self-reflection (Huisman et al., 2019) and stimulates deep-level text revisions (Scherer et al., 2024). FPA encourages students to articulate arguments, co-construct ideas, resolve conflicts, negotiate meanings, and formulate revision strategies (Bos and Tan, 2019). This offers learners opportunities to critically consider alternative arguments and engage in rational reasoning, thereby fostering their higher-order thinking in writing, such as critical thinking (Liu et al., 2023), reflective thinking (Cao et al., 2022), and metacognition (Zheng et al., 2018). During the FPA process, students receive feedback from multiple assessors, each with unique perspectives and insights, which enables personalized guidance for each student (Brkić et al.,

2024). Despite its effectiveness, FPA, as a form of human feedback, is not without its limitations. It can sometimes be inconsistent or biased due to varying skill levels, personal relationship or subjective interpretations (Chen et al., 2024). Additionally, delays in providing feedback can hinder students' ability to reflect on and revise their work in a timely manner, potentially impacting their learning process. Automated writing evaluation (AWE), by integrating artificial intelligence (AI) and natural language processing (NLP) techniques, can generate real-time comments and rich personalized feedback for learners (Barrot, 2023; Li, 2023). This makes AWE a useful supplement to human feedback (Yao et al., 2021). Studies have thus incorporated AWE into FPA, with findings indicating improvements in writing performance (Liu et al., 2023), growth mindset (Yao et al., 2021), and critical thinking (Liu et al., 2023).

Studies examining the combined effects of AWE on students' writing development in an AR-based FPA learning mode remain limited. We aimed to fill the gap by integrating AWE into an AR-based formative peer assessment approach (AR-AFPA) for writing education. Its effectiveness was explored through a quasi-experimental study, comparing it with one group employing formative peer assessment and teacher assessment in an AR environment (AR-TFPA). The effectiveness of AR-TFPA was established by comparing it with another group using the same assessments in a traditional lecture-based teaching mode (TFPA). The following sections present an overview of the current state of writing education, highlighting the benefits of integrating AR, FPA, and AWE into writing instruction.

1.1 The current state of writing education

Writing is a fundamental competence for participating in society (Nunes et al., 2022), and fostering students' writing proficiency is a primary goal of schooling from primary grades through college (Graham, 2019). However, many students worldwide often struggle to achieve this objective during their compulsory education (Graham, 2019). For instance, two-thirds of secondary school students in the United States fail to meet the expected writing standards for their grade level (National Center for Educational Statistics, 2012). Similarly, nearly half of sixth-grade students in Chile face difficulties with coherence, text organization, and idea development (Agencia de Calidad de la Educación, 2017). Previous research has identified several common challenges students may encounter in their efforts to become proficient writers: (a) difficulty retrieving related knowledge and experiences from long-term memory to organize and transform ideas into written text (Li et al., 2023), (b) inability to make deep-level text revision and self-reflection (Rietdijk et al., 2018), and (c) struggles with developing coherent thoughts and regulating cognitive processes (Teng & Yue, 2023). This highlights the need to change how writing is taught and learned.

Writing involves interactions among the environment, prior knowledge, memory, and reasoning (Lamb & Etopio, 2019). A visually intensive learning context is believed to activate prior knowledge and experiences stored in long-term memory (Li et al., 2023). Researchers have employed high-visualization tools (e.g., AR, VR) to stimulate learners' long-term memory related to writing topics, helping them

transcribe their inner thoughts into coherent sentences and complete texts (Chen et al., 2025b; Wang, 2017). Meanwhile, formative evaluation, which provides feedback on learners' written work or strategies, can enhance text revision and foster deep reflection (Graham, 2019; Rietdijk et al., 2018). As Graham (2022) noted, even younger learners who struggle with writing can improve their text quality through effective revision strategies. A meta-analysis by Scherer et al. (2024) revealed that combining different feedback forms (e.g., AWE, peer feedback, teacher feedback) is more beneficial for developing learners' writing skills than using either form alone. Moreover, learners must apply their knowledge, devise creative strategies, regulate their thinking, compare and question ideas, evaluate their progress, and make revisions (Teng & Yue, 2023). These processes not only enhance students' writing abilities but also cultivate critical thinking and thoughtful expression (Graham, 2022). Writing thus provides a conducive environment for fostering higher-order thinking (HOT), which, in turn, supports the development of writing skills (Wu et al., 2024).

HOT, which refers to cognitive abilities such as analysis, inferencing, generalization and problem solving in complex situations, plays a crucial role in developing writing proficiency (Zhan et al., 2023). According to thinking orientation, HOT can be divided into two categories: convergent and divergent. Convergent HOT (e.g., critical thinking and metacognition) is mainly guided by logical reasoning and adherence to established multiple criteria, while divergent HOT (e.g., creativity and problem solving) tends to be spontaneous, free-flowing and non-linear (Zhan et al., 2023). In this study, we investigated the effects of the AR-AFPA approach on developing writing skills by examining its impact on HOT. As a key determinant of students' learning outcomes (Teng & Yue, 2023), HOT has been a long-standing education objective (Lee & Choi, 2017). However, its role in writing education remains underexplored, creating a significant gap that this study aimed to address.

1.2 Benefits of AR for writing learning

Augmented reality (AR) is a technology that overlays virtual objects onto the real world to enhance users' experiences (Milgram et al., 1995). Due to its potential to create authentic contexts (Li et al., 2023), provide learning scaffolds (Wang, 2017), and boost motivation (Lin et al., 2022), AR has generated growing interest among researchers in applying it to support writing learning (Abdelrahim, 2023; Li et al., 2023; Wang, 2017). For example, Li et al. (2023) introduced a motivational augmented reality (MAR) approach to help elementary students learn Chinese writing. Their study found that MAR not only improved students' writing skills in feature description and creativity, but also sustained their writing motivation. Similarly, Wang (2017) employed AR to assist high school students in Chinese writing, leading to improvements in content control, structure and word choice. In another study, Allagui (2021) used AR to support college students' descriptive paragraph writing, finding that it evoked learners' cognitive skills in idea expression, text organization, and word choice. These applications demonstrated AR's effectiveness in terms of enhancing writing instruction by providing motivational learning experiences that help students overcome uncertainty in the writing process (Li et al., 2023). However,

AR does not address the limitations of the one-to-many teacher feedback mode, which often lacks personalized and instant feedback. These limitations prevent students from engaging in deep reflection on their writing, leading to superficial revision and less writing development (Liu et al., 2023; Scherer et al., 2024).

1.3 Benefits of FPA for writing learning

Formative peer assessment (FPA) is a strategy that helps mitigate the limitations of the sole use of AR in writing education. It refers to a process whereby learners assess their peers' work by providing scores or feedback that can be utilized to improve writing performance (Huisman et al., 2019; Topping, 1998). In FPA activities, students play dual roles: as assessors, they provide ratings and feedback, while as assesseees, they analyze the feedback received from their peers and revise their work accordingly (Cheng et al., 2015). This bidirectional process can activate a range of complex cognitive processes related to HOT, including comparison, feedback co-construction, reflective thinking, feedback evaluation, reflection, and cognitive regulation (Zhan et al., 2023). A meta-analysis of FPA's impact on HOT by Zhan et al. (2023) found that providing feedback simulates divergent HOT such as problem solving and creative thinking, while receiving feedback involves convergent HOT such as critical reflection, judgment, metacognition, and self-regulation. However, previous research on PFA in writing rarely emphasized its potential to cultivate both divergent and convergent HOT, with most studies primarily focusing on skills specific to convergent HOT. For instance, Liu et al. (2023) and Zou et al. (2023) reported a positive impact of FPA on fostering critical and reflective thinking in writing. Similarly, Zheng et al. (2018) demonstrated that discussion during FPA enhanced students' metacognitive awareness. The effectiveness of FPA in writing education has been explored by researchers, and it offers a solution to the problem of lack of personalized feedback (Scherer et al., 2024; Shulgina et al., 2024). FPA allows students to receive feedback from peers, who can provide more diverse perspectives and tailored insights (Huisman et al., 2019). Additionally, because peers are often at a similar learning level, their feedback tends to resonate more with writers' own understanding, highlighting areas that may be overlooked by teachers (Huisman et al., 2019). However, the effectiveness of peer feedback is influenced by students' skill levels, personal relationships, and subjective interpretations, which can occasionally result in biased or unaddressed feedback (e.g., erroneous use of vocabulary, grammatically wrong sentences) (Chen et al., 2024). Additionally, as a delayed feedback method, it cannot provide immediate responses, which may hinder students from revising their writing promptly.

1.4 Benefits of AWE for writing learning

Automated writing evaluation (AWE) can help address the challenges in FPA by providing unbiased feedback and reducing the human feedback burden (Fu et al., 2022). AWE refers to the capability of a computer technology to rate and assess written text (Nunes et al., 2022). Due to its prompt, rapid, frequent and intelligent

nature, AWE can provide students with feedback without delay, thereby alleviating the burden of human feedback (Fu et al., 2022). However, researchers have also cautioned about its drawbacks. For example, AWE often follows a one-size-fits-all approach and primarily identifies surface-level writing issues, while FPA tends to focus on deep-level aspects of writing (Li et al., 2015; Link et al., 2022). Thus, AWE should be viewed as a useful supplement to, rather than a replacement for, human feedback (i.e., teacher and peer feedback) (Cen & Zheng, 2024; Fu et al., 2022).

Some researchers have integrated AWE with FPA in writing instruction, demonstrating its effectiveness in terms of assessing surface-level aspects of writing issues and reducing the human feedback burden (Liu et al., 2023; Yao et al., 2021). For example, Yao et al. (2021) empirically found that AWE provides extensive surface-level corrective feedback (e.g., at the punctuation, word, and sentence levels), which complements FPA by identifying micro-level issues such as grammar and mechanics, thus cultivating students' writing motivation, language mindsets, and positive mentality. Similarly, Liu et al. (2023) verified the potential of this integration for enhancing students' writing performance, learning motivation, critical thinking and in-depth reflection. Meta-analytical evidence, however, remains inconclusive: a recent meta-analysis by Cen and Zheng (2024) discovered that although multiple-source feedback (e.g., AWE feedback combined with peer feedback, or teacher feedback combined with peer feedback) had a stronger motivational impact on writing learning outcomes compared to single-source feedback, the results of AWE combined with FPA should be interpreted with caution, as the number of effect sizes was too few to reach statistical significance. This highlights the need for further investigation into the effects of AWE combined with FPA, as well as comparisons with other multiple-source feedback programs, such as FPA combined with teacher assessment.

1.5 Types of feedback

Feedback, as a crucial component of evaluation practices, has been shown to promote revision behaviors and improve writing quality (Cen & Zheng, 2024; Scherer et al., 2024). It enables students to identify potential writing issues from a reader's perspective, thereby facilitating self-reflection and deep, substantive revisions of their written texts (Shulgina et al., 2024). Wu and Schunn (2023) suggested that students who engage in extensive revisions are more likely to produce higher-quality compositions.

The types of feedback students receive can affect their learning outcomes (Tan & Chen, 2022; Topping, 1998). Some researchers have analyzed the role of feedback types based on its affective and cognitive characteristics. For example, in an online learning context, Wu and Schunn (2020) classified feedback into six types: identification, explanation, suggestion, solution, mitigating praise, and hedges. Similarly, in an online assessment context, Cheng et al. (2015) categorized feedback into eight types: supporting, opposing, direct correction, personal opinion, guidance, evaluating, reflecting, and irrelevant. As for the impact of feedback types on learning outcomes, previous research has shown that cognitive feedback is particularly effective

for improving students' learning outcomes. Cheng et al. (2015) found that cognitive feedback was more beneficial than affective feedback for enhancing writing performance. However, research findings on the influence of affective feedback have been mixed. For example, Wu and Schunn (2020) observed that mitigating praise that acknowledges peers' writing can promote active text revision, thereby leading to improved writing performance. In contrast, Tan and Chen (2022) found that affective feedback had no significant effect on how students responded to it. In summary, different characteristics of feedback can result in varying levels of learning outcomes. Guided by the classification criteria of Tseng and Tsai (2007), this study examined the feedback received by students based on its affective and cognitive features (see Table 7). We aimed to examine how feedback affects learners' writing performance and higher-order thinking (HOT) across different learning approaches, with a focus on its mediating role.

1.6 The present study

To address the research gaps identified in the literature, this study integrating FPA and AWE into an AR environment (AR-AFPA), with the aim of examining its effectiveness in terms of supporting the writing process. Specifically, we compared the effects of TFPA with AR-TFPA, and AR-AFPA with AR-TFPA on writing performance. Moreover, cultivating HOT has long been a core goal in most educational practices (Lee & Choi, 2017). Previous studies on writing instruction have focused on developing single HOT skills, such as critical thinking (Abdelrahim, 2023). More recent studies, such as those by Wu et al. (2024), have explored how different writing strategies impact a broader range of HOT abilities, including critical thinking, creative thinking, and problem solving. Building on these findings, we investigated how the three proposed learning approaches influenced both convergent HOT (e.g., critical thinking and metacognition) and divergent HOT (e.g., problem-solving ability and creative thinking). Although prior research has examined how different pedagogical approaches affect the quantity of cognitive and affective feedback students received, as well as the direct effects of such feedback on writing performance and HOT (Cheng et al., 2015), the mechanisms by which these feedback types mediate the relationships between different learning approaches (i.e., TFPA vs. AT-TFPA; AR-TFPA vs. AR-AFPA), writing performance and HOT remain unclear. To explore these mechanisms, this study proposed a hypothetical mediation model based on previous studies (Shulgina et al., 2024; Wu et al., 2024) (Fig. 1). This model investigated how different types of feedback (i.e., affective and cognitive feedback) under the three approaches mediated the relationships between learning approaches and students' improvements in writing performance and HOT. This study aimed to answer the following research questions:

- (1) How do the proposed learning approaches (FPA vs. AR-TFPA; AR-TFPA vs. AR-AFPA) impact the development of writing performance?
- (2) How do the proposed learning approaches (FPA vs. AR-TFPA; AR-TFPA vs. AR-AFPA) impact the development of both convergent and divergent HOT?

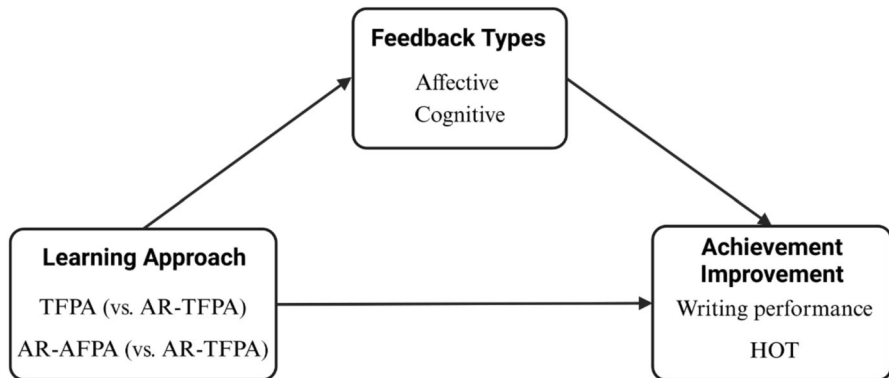


Fig. 1 Conceptual model

- (3) What are the mechanisms by which cognitive and affective feedback mediate the relationship between these learning approaches (FPA vs. AR-TFPA; AR-TFPA vs. AR-AFPA), writing performance and HOT?

2 Development of a writing learning system

Figure 2 illustrates the structure of the writing learning system, which consists of three modules: the AR module, the assessment module, and the management module. The AR module engages students by presenting learning materials and tasks in an immersive manner. The assessment module supports FPA activities, integrating either teacher feedback or AWE.

The management module oversees the design of FPA activities and monitors the learning process, allowing teachers to adapt their pedagogical strategies as needed.

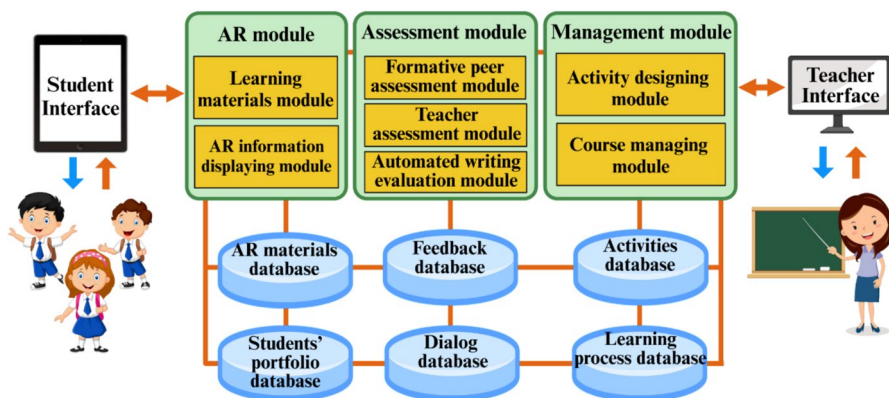


Fig. 2 The structure of the writing learning system

Additionally, the system includes several databases: an AR learning materials database, a students' portfolio database, a feedback database, a dialogue database, an activity database, and a learning process database that records students' learning behaviors.

This system adopts the peer assessment cycle model proposed by Kollar and Fischer (2010), which originally comprised four stages: task performance, feedback provision, feedback reception, and revision. We extended this model by introducing two additional stages: peer analysis and peer dialog, forming a six-stage FPA cycle (Fig. 3). Grounded in self-assessment theory, the stages of peer analysis, feedback provision, and feedback reception enable students to use peers' work as external reference points for self-assessment. This fosters students' reflection on their strengths and weaknesses, generates internal feedback, and encourages adjustments to their work and learning processes (Alemdag & Yildirim, 2022). From a constructivist perspective, peer dialog facilitates interpretive meaning construction by sharing interpretations, negotiating meanings, and clarifying expectations (Filius et al., 2018). As a result, both assessors and those being assessed engage in reflective knowledge construction within the FPA cycle. Upon completing the revision stage, this feedback loop closes, leading to enhanced student knowledge (Tan &

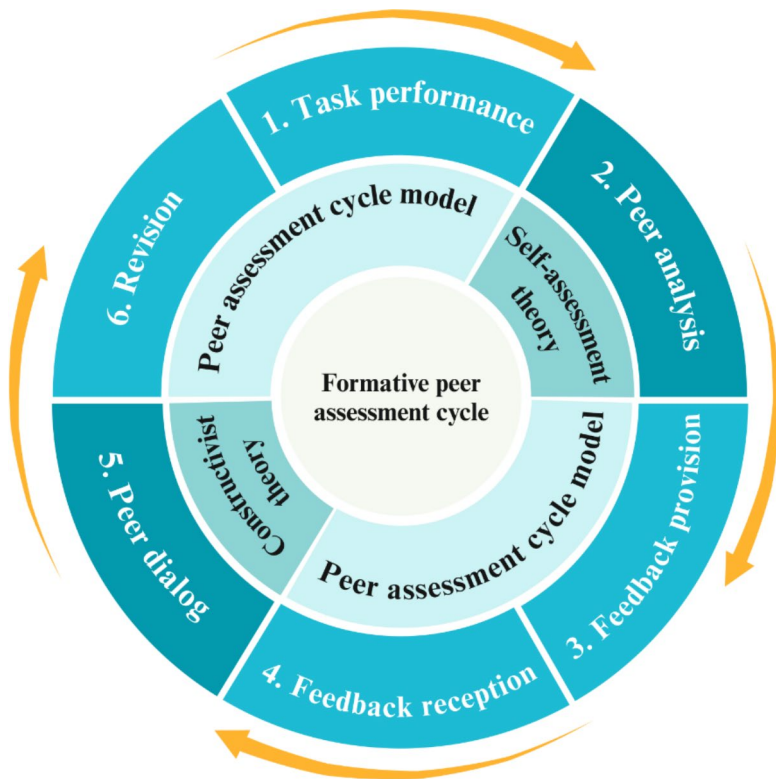


Fig. 3 Formative peer assessment cycle model

Chen, 2022). The following section will detail the specific activities involved in each of these six stages.

(1) Task performance

In this stage, students were assigned to write a narrative composition on the theme of “A Magical Adventure Journey,” a topic aligned with the requirements of The National Chinese Curriculum Standards for Chinese primary school (Ministry of Education, 2022). Before writing, students in the AR-TFPA and AR-AFPA groups used an AR application named “Explore Wild Animals” (see Fig. 4). By scanning a companion book with a mobile application on their smartphones or tablets, students explored virtual scenes enriched with audio and text prompts. These prompts scaffolded content learning by guiding interactions with virtual objects, such as a tiger seemingly leaping onto their hands, enhancing their sense of immersion. Students could also adjust the progress bar to revisit scenes, pause for detailed observation, and capture key moments with the screenshot function for later reference. These features supported detailed observation and helped students organize their ideas effectively for writing.

In contrast, students in the TFPA group used the printed materials (see Fig. 5) with the same content. After these preparatory activities, all three groups completed their writing assignments and submitted them to an online FPA platform developed using the open-source Workshop plugin of Moodle (2022) (version 3.11.11).

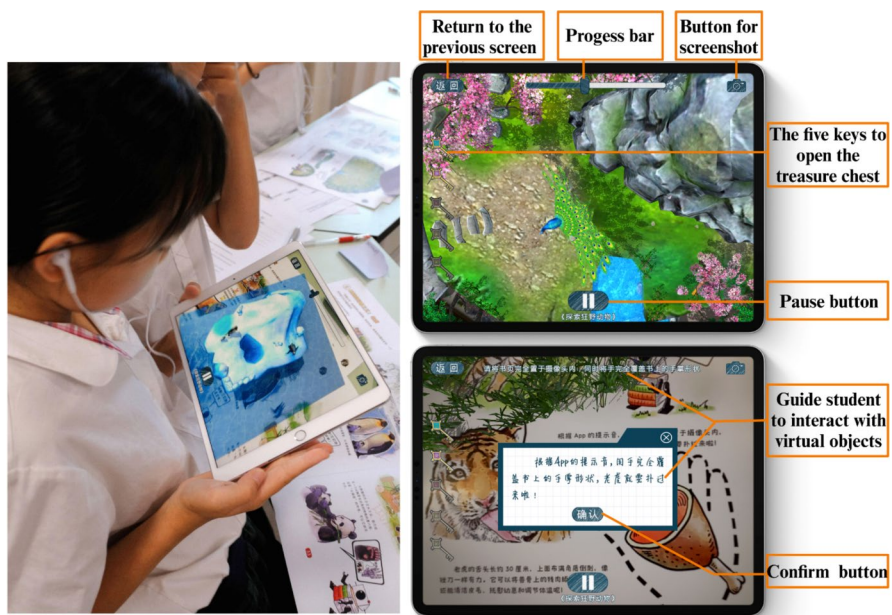


Fig. 4 Students observing and interacting with the AR system



Fig. 5 Illustration of students' course learning process

(2) Peer analysis

Teachers distributed student submissions from all three groups to their peers within their own group for evaluation. Figure 6a illustrates the peer assessment interface, where students reviewed and assessed their peers' work based on predefined criteria. This interface enabled students to compare their own work with that of their peers, reflect on their writing's strengths and weaknesses, and develop strategies for revision and writing process adjustments.

(3) Feedback provision

Students in all three groups provided scores and feedback on their peers' work based on predefined criteria. They scored and commented on the submissions across four dimensions: accuracy, organization, expression, and creativity (Fig. 6b). Prior to the feedback activity, the teacher explained and discussed these four criteria in class using sample writings. Subsequently, students practiced applying the criteria with two additional sample writings. Their feedback highlighted strengths, identified weaknesses, and offered constructive suggestions. To ensure anonymity, student names were replaced with numerical IDs, so assessors and those receiving feedback remained unaware of each other's identities.

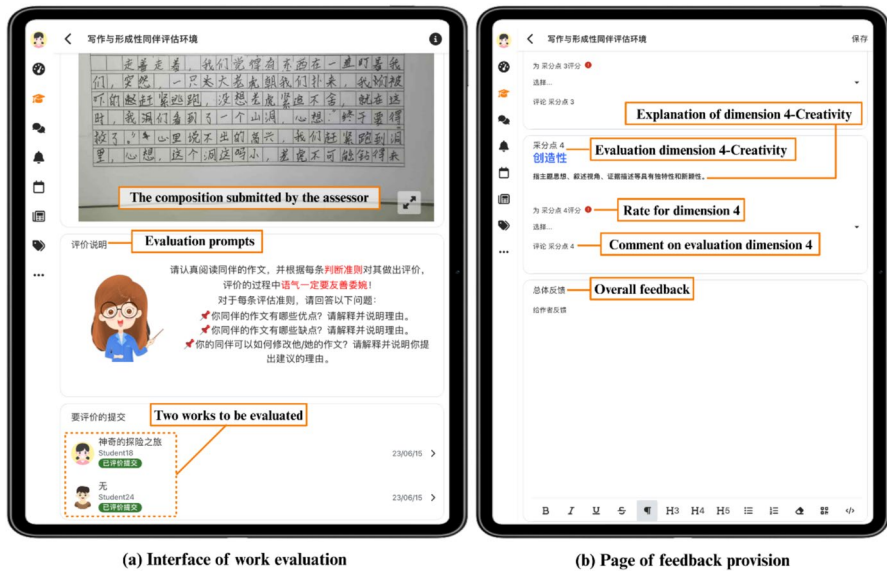


Fig. 6 Screenshot of the feedback provision module

(4) Feedback reception

In this phase, students in the TFPa and AR-TFPa groups received feedback in the form of scores and comments from two peers (Fig. 7a) and one teacher (Fig. 7b). Students in the AR-AFPA group received feedback from two peers (Fig. 7a) and an AWE system (the INCLASS intelligent educational platform, 2018) (Fig. 8). Students were then required to evaluate the clarity of the received feedback, assess their agreement with it, and reflect on their revision plans. By integrating AWE and FPA, students could compare and merge subjective peer feedback with objective AWE evaluations. In the subsequent peer dialog stage, they could then discuss and question the differences and similarities between them. This dynamic interaction fostered deeper reflection and a more critical engagement with peer feedback.

(5) Peer dialog

Students from all three groups engaged in discussions and analysis of the feedback in a forum format (Fig. 9). At the class level, they could share their thoughts, ask questions about the peer feedback, and participate in further discussions on the details of their work. This collaborative process helped students refine their revisions and improve their writing based on the received feedback.

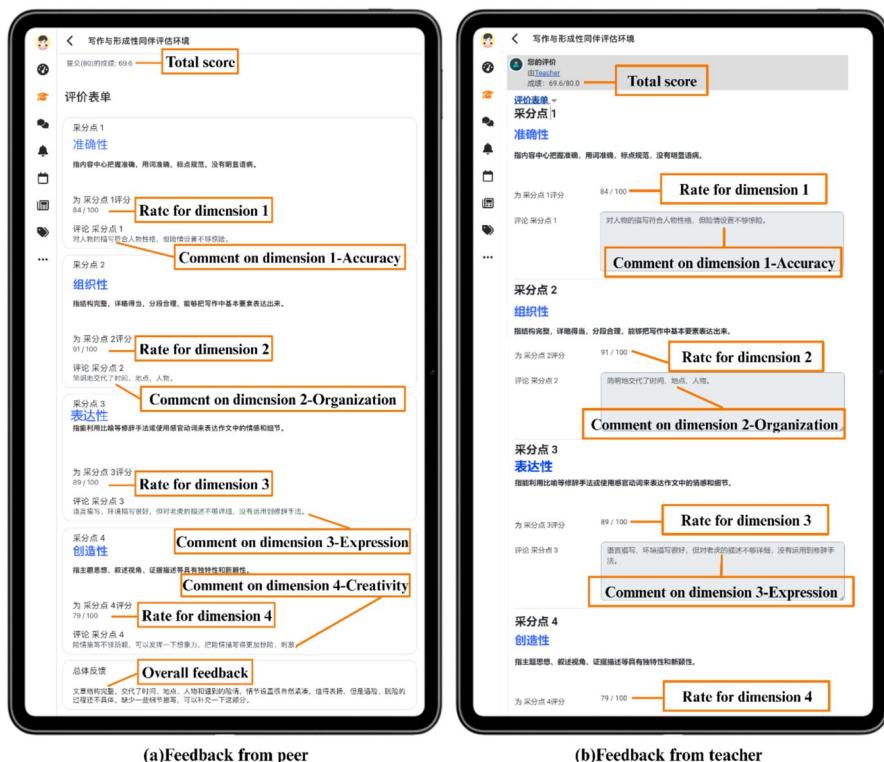


Fig. 7 Screenshot of the feedback reception module

(6) Revision

Finally, students in the TFPA and AR-TFPA groups revised their compositions using feedback from peers and teacher, while those in the AR-AFPA group utilized feedback from peers and AWE. All three groups then resubmitted their revised drafts to the online FPA platform.

3 Methods

3.1 Participants

A total of 137 fifth-grade students, aged 10 to 11 years on average, participated in this study. Participants were informed about the study's purpose and their right to withdraw at any time. Of the original 137 participants, 27 failed to complete the entire study process, resulting in 110 final subjects. These students were randomly



Fig. 8 Screenshot of the AWE feedback module

distributed across three groups: one group was designated as the TFPA group (24 females and 11 males), another as the AR-TFPA group (16 females and 21 males), and the remaining group as the AR-AFPA group (14 females and 24 males). To justify the sample size of this study, a post hoc power analysis was conducted using G*Power 3.1 (Faul et al., 2007). The analysis was based on an effect size of $f=0.35$, as suggested by Storbeck and Clore (2011), with an alpha level of 0.05 and two groups. The total sample sizes were 72 for the TFPA and AR-TFPA groups, and 75 for the AR-AFPA and AR-TFPA groups. The statistical power for the TFPA

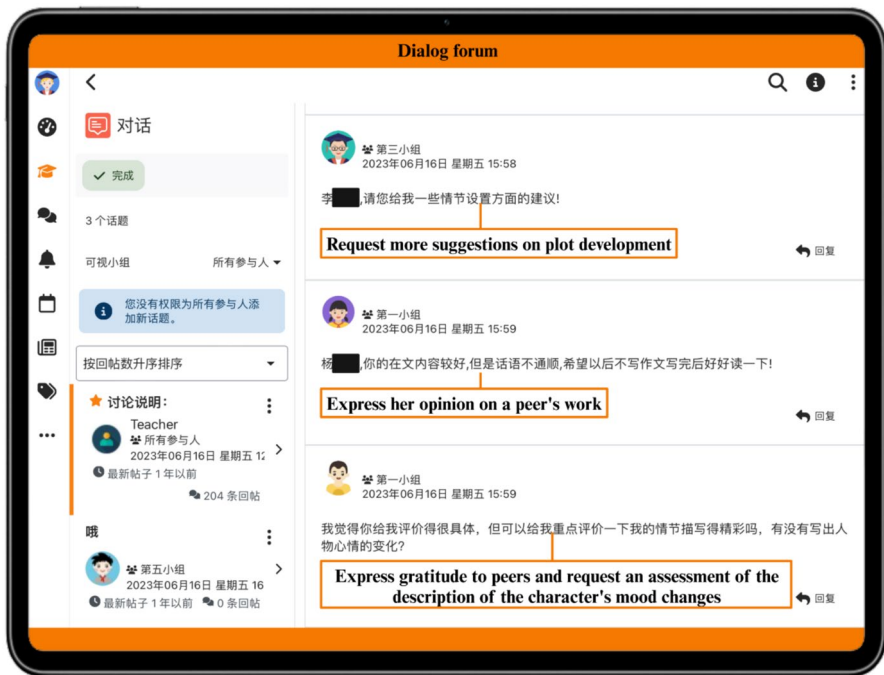


Fig. 9 Screenshot of the dialog forum

vs. AR-TFPA and AR-AFPA vs. AR-TFPA groups was 0.84 and 0.85, respectively, indicating that the sample size had sufficient power in the analysis of covariance (ANCOVA) to detect the main effects. Additionally, a Monte Carlo power analysis for indirect effects method was performed within R 4.2.3 (Schoemann et al., 2017), with a sample size of 110 and 5,000 replications. The statistical power values ranged from 0.81 to 0.98, suggesting that the sample size was adequate to detect the indirect effects in the parallel mediation models. To ensure consistency, all groups were taught by the same teacher.

3.2 Procedures

Figure 10 illustrates this study's experimental process, which comprised four phases: (1) Pretest, (2) Learning and writing, (3) Peer assessment, and (4) Posttest. The study spanned 5 weeks, with weekly 90-min classroom sessions. In the first phase, all three groups completed a pretest, including a writing task on the theme of "My Hometown" and a HOT questionnaire. From the second to the fourth week, students entered the learning and writing phase. The TFPA group utilized a paper material-based learning approach and submitted a draft composition on "A Magical Adventure Journey" by the end of the fourth week. In contrast, the AR-TFPA and AR-AFPA groups engaged in writing activities within an AR environment and submitted their drafts in the fourth week. In the fifth week, students participated

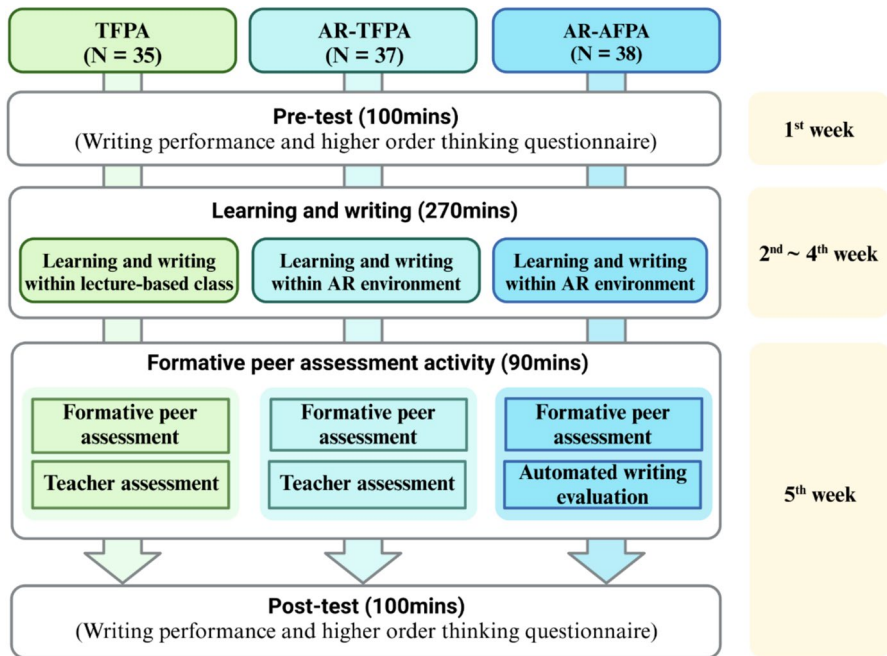


Fig. 10 Overview of the study design

in the FPA activity. All groups accessed an online FPA platform to read, rate, and comment on their peers' work. Students in the TFPA and AR-TFPA groups received additional feedback from a teacher to enhance their critical uptake of peer feedback, while the AR-AFPA group received additional feedback from an AWE system. Following the feedback, students revised and resubmitted their drafts: the TFPA and AR-TFPA groups revised theirs based on peer and teacher feedback, while the AR-AFPA group made revisions based on peer and AWE feedback. Finally, during the fifth week, all students completed a posttest, which included a HOT questionnaire and their revised writings.

3.3 Measurements

3.3.1 The rubric of Chinese writing performance

The rubric for assessing Chinese writing performance was adapted from the Chinese descriptive writing assessment scale developed by Yang et al. (2021) and was refined based on suggestions from Chinese language teachers. Table 6 presents the rubric's four dimensions: accuracy, organization, expression, and creativity. These dimensions evaluate whether learners' writing meets the prescribed requirements, includes all essential elements, effectively employs rhetorical devices and sensory

verbs to convey emotions, and presents unique, innovative perspectives. Together, these criteria provide a comprehensive evaluation of pupils' Chinese narrative writing. Each composition was evaluated on a 100-point scale, with 25 points allocated to each dimension. Two experienced Chinese language teachers conducted the evaluations in a double-blind manner using this rubric. The intra-class correlation coefficient (ICC) between the two raters was 0.901, demonstrating an excellent level of reliability (Koo & Li, 2016). Therefore, the average score given by the two teachers was adopted as the final grade for students' writing performance.

3.3.2 Higher-order thinking questionnaire

The HOT questionnaire, adapted from Lee and Choi (2017), comprised 16 self-report items across four dimensions: problem solving (3 items), creative thinking (3 items), metacognition (4 items), and critical thinking (6 items) (see Table 7). Problem solving and creative thinking were categorized as divergent HOT, while metacognition and critical thinking were classified as convergent HOT. HOT was measured using a 5-point Likert scale (1 = *strongly disagree*, 5 = *strongly agree*). All four dimensions were valid and reliable measures for HOT (see Table 8). Cronbach's alpha (α) and McDonald's Omega (ω) coefficients revealed acceptable internal consistency and reliability for the critical thinking ($\alpha = 0.855$, $\omega = 0.856$), metacognition ($\alpha = 0.877$, $\omega = 0.878$), creativity ($\alpha = 0.862$, $\omega = 0.862$), and problem solving ($\alpha = 0.796$, $\omega = 0.802$) dimensions. To evaluate construct validity, an Exploratory Factor Analysis (EFA) using Principal Component Analysis (PCA) with Varimax rotation was performed in IBM SPSS 26.0. The Kaiser–Meyer–Olkin (KMO) value was 0.879 and Bartlett's Test of Sphericity yielded a significant result (approx. $\chi^2 = 1063.24$, $p < 0.001$), confirming the rationale for conducting EFA. The EFA identified four factors with eigenvalues greater than 1, accounting for 71.37% of the total items. Subsequently, a Confirmatory Factor Analysis (CFA) was conducted using Amos 29.0, which indicated a good fit ($\chi^2(95) = 1.362$, $p < 0.001$, $TLI = 0.957$, $CFI = 0.967$, $GFI = 0.883$, $RMSEA = 0.058$, $SRMR = 0.056$) (Hu & Bentler, 1999). For all four dimensions, composite reliability (CR) values exceeded 0.7, and average variance extracted (AVE) values were above 0.5, suggesting the acceptable construct validity of the measurement items (Fornell & Larcker, 1981) (see Table 8).

3.3.3 Coding scheme of feedback

To analyze feedback from both the affective and cognitive dimensions, we adapted the coding frameworks developed by Tseng and Tsai (2007) (see Table 9). Affective feedback was categorized into three types: (A1) positive, (A2) negative and (A3) neutral. Positive feedback, including expression of support or praise, was coded as A1. Negative feedback, which includes criticism or negative emotions, was coded as A2. Feedback that neither praises nor criticizes was coded as A3.

Cognitive feedback was classified into five types: (C1) corrective, (C2) reinforcing, (C3) didactic, (C4) suggestive, and (C5) irrelevant. Corrective feedback, which identifies errors or provides direct corrections, was coded as C1. Reinforcing

feedback, expressing approval or positive feelings, was coded as C2. Didactic feedback, which includes lengthy explanations or lectures, was coded as C3. Suggestive feedback, offering guidance or hints for improvement, was coded as C4. Feedback unrelated to the content of writing was coded as C5. During coding, some feedback messages fell into multiple categories. For instance, feedback such as, “Overall, well-written, but I think adding more descriptions of the adventure would be better” includes both reinforcing feedback (“Overall, well-written”) and suggestive feedback (“I think adding more description of the adventure would be better”). In such cases, the feedback was coded as both C2 and C4. Two trained coders independently coded the feedback, achieving a Cohen’s kappa value of 0.86 ($p < 0.001$), indicating high consistency between coders. For instances of inconsistent coding, the two coders discussed and negotiated the discrepancies to reach a consensus.

3.4 Data analysis

We employed IBM SPSS 26.0 to perform analysis of covariance (ANCOVA) tests, examining differences in writing performance and HOT between the TFPA group and the AR-TFPA group, as well as between the AR-AFPA group and the AR-TFPA group. Pretest scores for writing performance and HOT served as covariates, with posttest scores serving as dependent variables.

To investigate the relationships between the feedback received under different learning approaches and improvements in writing performance and HOT before and after FPA, we conducted Chi-square tests, correlation analyses, and mediation analysis using Mplus 8.3 (Muthén & Muthén, 1998–2017). The mediation model employed a multiple parallel mediation model with a multi-categorical independent variable (Preacher & Hayes, 2008). In this model, the learning approach was the independent variable (dummy coded, with the AR-TFPA approach as the reference category), feedback types (A1-A3 and C1-C5) were the mediators, and improvements in writing performance and HOT were the dependent variables. Control variables included students’ pretest scores in writing and HOT, as well as gender and age. Mediation results were reported in terms of relative total effects, direct effects, and indirect effects (Fig. 12 and Table 5–6). We tested the significance of indirect effects using the Bootstrapping method, which is considered more effective than the Sobel test and the causal steps approach (Hayes, 2009). An indirect effect was deemed significant if the 95% bias-corrected bootstrap confidence interval, based on 5,000 bootstrap samples, did not include zero (Hayes, 2009).

4 Results

4.1 Writing performance

ANCOVA (Analysis of Covariance) was conducted to examine the impact of different learning approaches on students’ writing performance across total scores and the four dimensions. Levene’s test indicated no significant differences

between groups for total scores ($F_{(2, 107)} = 2.745$, $p = 0.069 > 0.05$), or for the accuracy ($F_{(2, 107)} = 0.062$, $p = 0.940 > 0.05$), organization ($F_{(2, 107)} = 1.234$, $p = 0.295 > 0.05$), expression ($F_{(2, 107)} = 1.505$, $p = 0.225 > 0.05$) or creativity ($F_{(2, 107)} = 2.861$, $p = 0.062 > 0.05$) dimensions. Additionally, the assumption of the homogeneity regression slopes was not violated for total scores or for the four dimensions with $F_{(2, 107)} = 0.894$ ($p = 0.412 > 0.05$), $F_{(2, 107)} = 1.783$ ($p = 0.173 > 0.05$), $F_{(2, 107)} = 0.831$ ($p = 0.439 > 0.05$), $F_{(2, 107)} = 1.167$ ($p = 0.315 > 0.05$), $F_{(2, 107)} = 0.143$ ($p = 0.867 > 0.05$), indicating that it was reasonable to perform ANCOVA.

As shown in Table 1, the ANCOVA results for posttest writing performance revealed significant differences among the three groups in total scores ($F_{(2, 106)} = 25.869$, $p < 0.001$), accuracy ($F_{(2, 106)} = 27.943$, $p < 0.001$), organization ($F_{(2, 106)} = 29.126$, $p < 0.001$), expression ($F_{(2, 106)} = 16.940$, $p < 0.001$) and creativity ($F_{(2, 106)} = 20.880$, $p < 0.001$). That is, after eliminating the influence of covariance (pretest) on the dependent variable (posttest), posttest writing performance were significantly different in the three groups. Specifically, the AR-AFPA group significantly outperformed the AR-TFPA group in total scores and organization ($F_{(1, 72)} = 4.731$, $p < 0.05$; $F_{(1, 72)} = 7.091$, $p < 0.05$), while the AR-TFPA group significantly exceeded the TFPA group ($F_{(1, 69)} = 25.030$, $p < 0.001$; $F_{(1, 69)} = 25.930$, $p < 0.001$). In terms of accuracy, the AR-AFPA group significantly surpassed the AR-TFPA group ($F_{(1, 72)} = 30.274$, $p < 0.001$), while the AR-TFPA group did not score statistically higher than the TFPA group ($F_{(1, 69)} = 2.597$, $p > 0.05$). Regarding expression, although the AR-AFPA group scored higher than the AR-TFPA group, the difference was not statically significant

Table 1 ANCOVA results of posttest writing performance: AR-TFPA vs. TFPA, and AR-AFPA vs. AR-TFPA learning approach

	Group	<i>N</i>	Mean	<i>SD</i>	Adj.M	<i>SE</i>	<i>F</i>	Pairwise comparisons
Total	AR-AFPA	38	85.09	7.05	84.93	1.08	25.869***	AR-AFPA > AR-TFPA* AR-TFPA > TFPA***
	AR-TFPA	37	81.41	7.97	81.72	0.94		
	TFPA	35	75.31	5.53	74.98	0.97		
Accuracy	AR-AFPA	38	86.21	6.33	86.40	1.05	27.943***	AR-AFPA > AR-TFPA***
	AR-TFPA	37	78.45	7.86	78.60	1.05		
	TFPA	35	76.33	6.41	76.16	1.08		
Organization	AR-AFPA	38	87.84	8.90	87.82	1.29	29.126***	AR-AFPA > AR-TFPA* AR-TFPA > TFPA***
	AR-TFPA	37	82.93	8.40	83.09	1.13		
	TFPA	35	74.97	6.34	74.80	1.17		
Expression	AR-AFPA	38	89.42	7.08	89.24	1.18	16.940***	AR-TFPA > TFPA***
	AR-TFPA	37	86.22	9.15	86.76	1.06		
	TFPA	35	81.33	7.04	80.76	1.09		
Creativity	AR-AFPA	38	76.89	7.44	76.50	1.24	20.880***	AR-TFPA > TFPA***
	AR-TFPA	37	78.03	8.32	78.27	1.08		
	TFPA	35	68.61	5.77	68.36	1.12		

*N*Number of participants, *SD*Standard deviation, *Adj.M*Adjusted means, *SE*Standard error. $p^* < 0.05$; $p^{***} < 0.001$

($F_{(1, 72)} = 2.955$, $p > 0.05$), whereas the AR-TFPA group significantly outperformed the TFPA group ($F_{(1, 69)} = 15.554$, $p < 0.001$). Notably, for creativity, the AR-AFPA group (mean = 76.89) scored lower than the AR-TFPA group (mean = 78.03) but higher than the TFPA group (mean = 68.61), with the AR-TFPA group significantly outperforming the TFPA group ($F_{(1, 69)} = 40.475$, $p < 0.001$). Therefore, it can be concluded that the AR-AFPA approach is more helpful for improving students' writing performance in terms of their total scores, accuracy, organization and expression, while the AR-TFPA approach appears more beneficial for fostering writing creativity.

4.2 Higher-order thinking

ANCOVA was employed to evaluate the effects of different learning approaches on students' HOT across overall HOT and its four dimensions. Levene's test indicated no significant differences between groups for overall HOT, critical thinking, metacognition, creativity or problem solving with $F_{(2, 107)} = 0.126$ ($p = 0.725 > 0.05$), $F_{(2, 107)} = 0.472$ ($p = 0.625 > 0.05$), $F_{(2, 107)} = 0.521$ ($p = 0.595 > 0.05$), $F_{(2, 107)} = 1.885$ ($p = 0.157 > 0.05$), and $F_{(2, 107)} = 1.172$ ($p = 0.314 > 0.05$), respectively. The assumption of the homogeneity regression slopes was not violated for overall HOT or for any of the four dimensions, with $F_{(2, 107)} = 0.856$ ($p = 0.428 > 0.05$), $F_{(2, 107)} = 0.121$ ($p = 0.886 > 0.05$), $F_{(2, 107)} = 0.241$ ($p = 0.786 > 0.05$), $F_{(2, 107)} = 0.021$ ($p =$

Table 2 ANCOVA results of posttest HOT: AR-TFPA vs. TFPA, and AR-AFPA vs. AR-TFPA learning approach

	Group	<i>N</i>	Mean	<i>SD</i>	Adj.M	<i>SE</i>	<i>F</i>	Pairwise comparisons
Overall HOT	AR-AFPA	38	16.32	1.90	16.43	0.35	5.718**	AR-TFPA > TFPA*
	AR-TFPA	37	16.06	2.68	16.21	0.42		
	TFPA	35	14.91	2.68	14.75	0.43		
Convergent HOT								
Critical thinking	AR-AFPA	38	4.21	0.69	4.23	0.11	4.377*	
	AR-TFPA	37	4.06	0.70	4.07	0.12		
	TFPA	35	3.76	0.74	3.76	0.12		
Metacognition	AR-AFPA	38	4.31	0.76	4.31	0.12	4.405*	
	AR-TFPA	37	4.03	0.77	4.07	0.13		
	TFPA	35	3.84	0.82	3.79	0.14		
Divergent HOT								
Creativity	AR-AFPA	38	3.82	0.94	3.85	0.16	0.622	
	AR-TFPA	37	3.81	1.11	3.81	0.16		
	TFPA	35	3.63	0.91	3.63	0.17		
Problem solving	AR-AFPA	38	3.98	0.78	3.99	0.13	4.752*	AR-TFPA > TFPA**
	AR-TFPA	37	4.16	0.89	4.20	0.13		
	TFPA	35	3.69	0.73	3.64	0.13		

NNumber of participants, SDStandard deviation, Adj.MAdjusted means, SEStandard error. $p^* < 0.05$; $p^{**} < 0.01$

$0.979 > 0.05$), $F_{(2, 107)} = 0.083$ ($p = 0.920 > 0.05$), respectively, confirming that it was appropriate to conduct ANCOVA.

As presented in Table 2, the ANCOVA results for posttest HOT showed significant differences among the three groups for overall HOT ($F_{(2, 106)} = 5.718$, $p = 0.004 < 0.01$), as well as for the critical thinking and metacognition dimensions of convergent HOT ($F_{(2, 106)} = 4.377$, $p = 0.015 < 0.05$; $F_{(2, 106)} = 4.405$, $p = 0.015 < 0.05$), and the problem solving dimension of divergent HOT ($F_{(2, 106)} = 4.752$, $p = 0.011 < 0.05$). However, no significant differences were found in the creativity dimension of divergent HOT ($F_{(2, 106)} = 3.262$, $p = 0.539 > 0.05$). In detail, the AR-AFPA group outperformed the AR-TFPA group in overall HOT, but the difference was not statically significant ($F_{(1, 72)} = 0.454$, $p > 0.05$), while the AR-TFPA group significantly outperformed the TFPA group ($F_{(1, 69)} = 5.868$, $p < 0.05$). For the problem solving dimension of divergent HOT, the AR-AFPA group (mean = 3.98) scored lower than the AR-TFPA group (mean = 4.16) but higher than the TFPA group (mean = 3.69), with the AR-TFPA group significantly outperforming the TFPA group ($F_{(1, 69)} = 8.836$, $p < 0.01$). No significant differences were observed in the critical thinking or metacognition dimensions of convergent HOT, either between the AR-AFPA and AR-TFPA groups ($F_{(1, 72)} = 1.321$, $p > 0.05$; $F_{(1, 72)} = 2.573$, $p > 0.05$) or between the AR-TFPA and TFPA groups ($F_{(1, 69)} = 3.262$, $p > 0.05$; $F_{(1, 69)} = 2.166$, $p > 0.05$). Thus, it can be concluded that the AR-AFPA approach is more effective in terms of fostering students'

Table 3 Chi-square test for received feedback between groups

Category	Type	Value	Group			χ^2	<i>p-value</i>
			TFPA	AR-TFPA	AR-AFPA		
Total		Count	227	272	504		
		Percentage (%)	100	100	100		
Affective	Positive	Count	127	118	156	68.63	< 0.001***
		Percentage (%)	55.9	43.4	31.0		
	Negative	Count	39	90	119		
		Percentage (%)	17.2	33.1	23.6		
	Neutral	Count	61	64	229		
		Percentage (%)	26.9	23.5	45.4		
Cognitive	Corrective	Count	55	103	178	64.47	< 0.001***
		Percentage (%)	24.2	37.9	35.3		
	Reinforcing	Count	71	64	85		
		Percentage (%)	31.3	23.5	16.9		
	Didactic	Count	18	8	40		
		Percentage (%)	7.9	2.9	7.9		
	Suggestive	Count	69	95	200		
		Percentage (%)	30.4	34.9	39.7		
	Irrelevant	Count	14	2	1		
		Percentage (%)	6.2	0.7	0.2		

$p^{***} < 0.001$

convergent HOT in writing, although it may hinder the development of divergent HOT, particularly problem solving.

4.3 The mediating effect of feedback

Table 3 shows the distribution of different types of affective and cognitive feedback received by students across the three learning approaches. The Chi-square test revealed a significant association between the distribution of both affective and cognitive feedback and the learning approaches ($\chi^2 = 68.63$, $p < 0.001$; $\chi^2 = 64.47$, $p < 0.001$). Pairwise post hoc tests further identified significant differences between the AR-TFPA group (2) and both the TFPA group (1) and the AR-AFPA group (3). Specifically, for affective feedback, the TFPA group (Bonferroni adjusted $p = 2.43\text{e-}4^{***}$) and AR-AFPA group (Bonferroni adjusted $p = 1.4664\text{e-}8^{***}$) both showed significant differences from the AR-TFPA group in the proportions of feedback types. Similarly, for cognitive feedback, significant differences were observed between the AR-TFPA group and both the TFPA group (Bonferroni adjusted $p = 1.2\text{e-}5^{***}$) and AR-AFPA group (Bonferroni adjusted $p = 5.528\text{e-}3^{**}$).

To explore the relationships between students' improvements in writing performance, HOT before and after FPA, and the feedback received, we first analyzed the potential correlations among these variables. Figure 11 illustrates the bivariate correlations between affective feedback (a) and cognitive feedback (b) with improvements in both writing performance and HOT. As shown in Fig. 11, writing performance improvement was significantly positively correlated with the amounts of neutral ($r = 0.47$, $p < 0.001$), positive ($r = 0.57$, $p < 0.001$), corrective ($r = 0.46$, $p < 0.001$), and suggestive ($r = 0.59$, $p < 0.001$) feedback, while it was significantly negatively correlated with negative ($r = -0.19$, $p < 0.05$) and irrelevant ($r = -0.20$, $p < 0.05$) feedback. Similarly, HOT improvement was significantly positively correlated with positive ($r = 0.30$, $p < 0.001$), neutral ($r = 0.60$, $p < 0.001$), corrective ($r = 0.51$, $p < 0.001$) and suggestive ($r = 0.62$, $p < 0.001$) feedback.

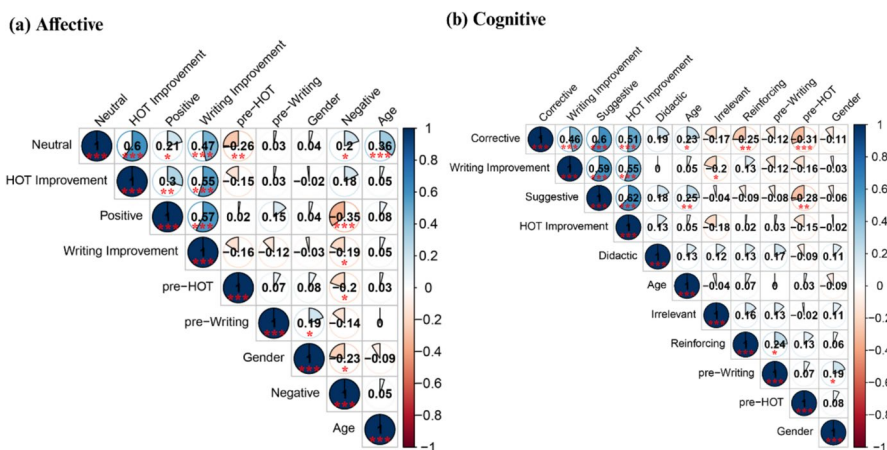


Fig. 11 Bivariate correlations between the main variables

0.001) feedback. Additionally, a positive correlation was observed between improvements in writing performance and HOT ($r = 0.55, p < 0.001$). These findings indicate significant correlations between the improvements in writing performance and HOT, and the amounts of received affective and cognitive feedback, suggesting the potential for further mediation analysis.

To examine whether the learning approach influences improvements in writing performance and HOT through received feedback, we constructed four parallel multiple mediator models. These models included the learning approach as the explanatory variable, with age, gender, and pretest scores for writing performance and HOT as the control variable. Since the explanatory variable was categorical with a total of three categories, two dummy variables were used to represent these categories, with the AR-TFPA learning approach serving as the reference category. All significant direct paths are depicted in Fig. 12; however, it does not show the relationships between the control variables and mediators, or between the control variables and the dependent variable, although these were accounted for in the calculated models. For the multi-categorical independent variable, two

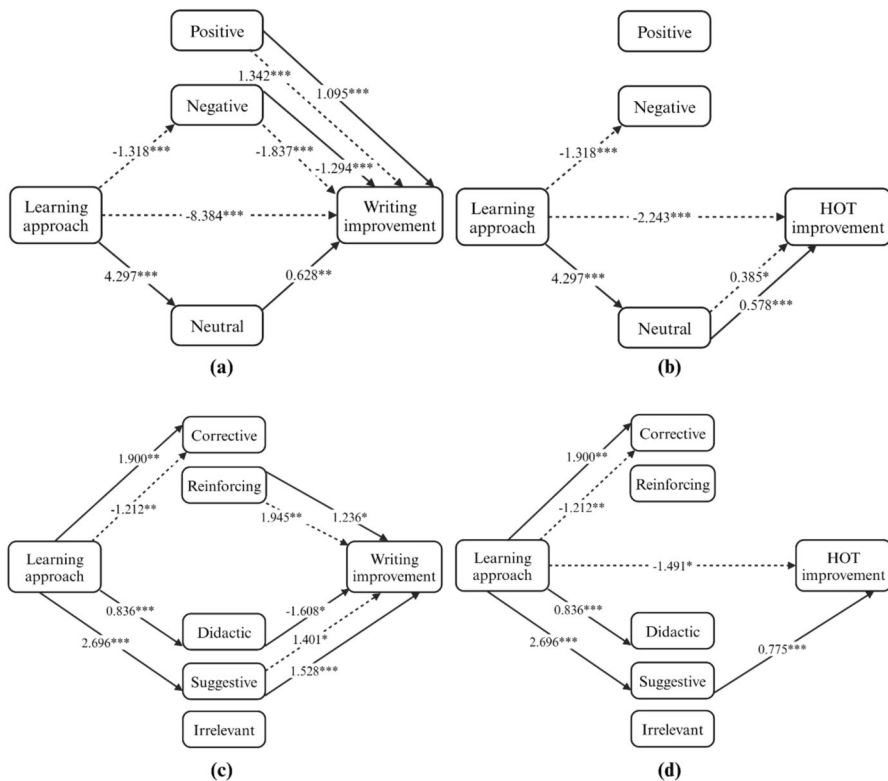


Fig. 12 Mediation analysis of the AR-AFPA (solid arrows) and the TFPA (dashed arrows) for writing performance and HOT improvement. Note. All parameter estimates are unstandardized. Only significant direct paths are displayed. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 4 Relative indirect effect for writing performance improvement

Paths (IV → MV → DV)	TFPA ^a				AR- AFPA ^a			
	Effect	SE	95% LLCI	95% ULCI	Effect	SE	95% LLCI	95% ULCI
Total	−5.37**	1.74	−8.696	−1.888	4.85*	2.10	0.685	9.008
LA → WI ^b	3.01*	1.30	0.484	5.620	2.80*	1.46	0.002	5.701
LA → A1 → WI	0.59	0.85	−0.929	2.440	1.00	0.73	−0.140	2.732
LA → A2 → WI	2.42**	0.75	1.164	4.151	−0.90	0.61	−2.252	0.163
LA → A3 → WI	0.01	0.19	−0.473	0.377	2.70**	0.86	1.081	4.543
Total	−5.40*	2.11	−9.206	−1.028	3.24*	2.30	−1.170	7.847
LA → WI ^b	−1.03	1.48	−4.028	1.832	4.21**	1.36	1.933	7.288
LA → C1 → WI	0.11	0.82	−1.455	1.841	0.80	0.65	−0.279	2.305
LA → C2 → WI	0.63	0.65	−0.326	2.554	0.66	0.46	−0.013	1.921
LA → C3 → WI	−0.45	0.57	−1.861	0.440	−1.35*	0.69	−2.903	−0.148
LA → C4 → WI	−0.84	0.84	−2.899	0.277	4.12***	1.15	2.224	6.857
LA → C5 → WI	−0.49	0.64	−1.983	0.549	−0.03	0.20	−0.457	0.203

^a Compared with the AR–TFPA learning approach; ^b total indirect effect; Bootstrap sample size = 5,000; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Abbreviations: LA Learning approach, A1 Positive feedback, A2 Negative feedback, A3 Neutral feedback, C1 Corrective feedback, C2 Reinforcing feedback, C3 Didactic feedback, C4 Suggestive feedback, C5 Irrelevant feedback

Table 5 Relative indirect effect for HOT improvement

Paths (IV → MV → DV)	TFPA ^a				AR- AFPA ^a			
	Effect	SE	95% LLCI	95% ULCI	Effect	SE	95% LLCI	95% ULCI
Total	−2.23**	0.67	−3.489	−0.866	3.00**	1.07	1.002	5.176
LA → HOTI ^b	0.01	0.32	−0.669	0.614	2.63***	0.64	1.432	3.949
LA → A1 → HOTI	0.07	0.15	−0.069	0.581	0.14	0.18	−0.040	0.701
LA → A2 → HOTI	−0.06	0.26	−0.646	0.408	0.01	0.16	−0.290	0.386
LA → A3 → HOTI	0.01	0.15	−0.345	0.277	2.48***	0.59	1.491	3.843
Total	−2.09**	0.76	−3.468	−0.465	2.60*	1.17	0.416	4.986
LA → HOTI ^b	−0.60	0.35	−1.419	0.000	2.40**	0.91	0.734	4.132
LA → C1 → HOTI	−0.19	0.28	−0.846	0.298	0.26	0.32	−0.273	1.001
LA → C2 → HOTI	0.07	0.14	−0.064	0.591	0.05	0.18	−0.203	0.496
LA → C3 → HOTI	−0.08	0.17	−0.539	0.197	0.01	0.38	−0.708	0.809
LA → C4 → HOTI	−0.23	0.21	−0.756	0.070	2.09**	0.76	0.800	3.718
LA → C5 → HOTI	−0.18	0.17	−0.589	0.078	−0.01	0.07	−0.232	0.048

^a Compared with the AR–TFPA learning approach; ^b total indirect effect; Bootstrap sample size = 5,000; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Abbreviations: LA Learning approach, A1 Positive feedback, A2 Negative feedback, A3 Neutral feedback, C1 Corrective feedback, C2 Reinforcing feedback, C3 Didactic feedback, C4 Suggestive feedback, C5 Irrelevant feedback

relative direct effects for each dependent variable and two relative indirect effects for each mediator are reported (see Table 4 and 5).

Table 4 illustrates the mediating role of affective and cognitive feedback between learning approaches and improvements in writing performance. For the TFPA (vs. AR-TFPA) approach, the total indirect effect of affective feedback on writing performance improvement was significantly positive ($\beta = 3.01$, 95% CI [0.484, 5.620]), while the direct effect was significantly negative ($\beta = -8.384$, $p < 0.001$) (see Fig. 12a), indicating a suppression effect (Preacher & Hayes, 2008). This suggests that negative feedback is a suppressor in the relationship between the TFPA (vs. AR-TFPA) approach and writing performance improvement. Notably, none of the cognitive feedback mediators showed significant mediation effects. For the AR-AFPA (vs. AR-TFPA) approach, both the total indirect effects of affective and cognitive feedback were significantly positive ($\beta = 2.80$, 95% CI [0.002, 5.701]; $\beta = 4.21$, 95% CI [1.933, 7.288]). Specifically, the amount of neutral feedback mediated the relationship between the AR-AFPA (vs. AR-TFPA) approach and writing performance improvement ($\beta = 2.70$, 95% CI [1.081, 4.543]). Additionally, the amounts of didactic and suggestive feedback mediated the relationship between the AR-AFPA (vs. AR-TFPA) approach and writing performance improvement ($\beta = -1.35$, 95% CI [-2.903, -0.148]; $\beta = 4.12$, 95% CI [2.224, 6.857]).

Table 5 illustrates the mediating role of affective and cognitive feedback between learning approaches and HOT improvement. For the TFPA (vs. AR-TFPA) approach, none of the affective or cognitive feedback mediators showed significant indirect effects. In contrast, for the AR-AFPA (vs. AR-TFPA) approach, both affective and cognitive feedback demonstrated significant positive total indirect effects ($\beta = 2.63$, 95% CI [1.432, 3.949]; $\beta = 2.40$, 95% CI [0.734, 4.132]). Specifically, the amount of neutral feedback mediated the relationship between the AR-AFPA (vs. AR-TFPA) approach and HOT improvement ($\beta = 2.48$, 95% CI [1.491, 3.843]), while suggestive feedback mediated the relationship between the AR-AFPA (vs. AR-TFPA) approach and HOT improvement ($\beta = 2.09$, 95% CI [0.800, 3.718]).

5 Discussion

5.1 Effect of learning approaches on writing achievement

The study findings indicated that students who adopted the AR-TFPA learning approach significantly outperformed those using the TFPA approach on total writing scores, as well as on the dimensions of organization, expression, and creativity. This aligns with previous research which suggested that AR can enhance students' writing achievement (Abdelrahim, 2023; Allagui, 2021). First, by overlaying virtual scenarios and 3D models onto physical environments, AR technology transforms abstract concepts into concrete visual representations (Li et al., 2023). This visual enhancement may deepen learners' understanding of writing themes and help them clarify narrative sequences and logical structures. Consequently, learners can more effectively organize information, structure their paragraphs, and ultimately produce complete compositions. Second, AR's dynamic linguistic scaffolding, which

integrates contextual textual prompts with synchronized audio cues, may enable students to express emotions and narrate storylines with suitable vocabulary and sentences, thereby enriching their language expression (Wang, 2017). Third, AR's immersive, interactive and navigational features enhance students' motivation to learn and foster creative thinking (Wei et al., 2015). These features allow AR to simulate various scenarios and story settings while providing real-time feedback to respond to students' interactions. This exposure to varied narrative elements may enrich students' creative resources and inspire unique writing perspectives, making writing more personalized and creative. However, our study found no significant difference between the two groups in the accuracy dimension of writing performance. As suggested in Hayes's (1996) writing process theory, one possible explanation is that AR-based learning activities demand high-level cognitive processing. Due to the limited capacity of working memory, students might struggle to handle all cognitive tasks required in writing, which may lead to neglect of details such as vocabulary quality and punctuation. Another explanation might be the novelty of AR technology (Li et al., 2023). This novelty effect may attract students' focus on the exploration of the technology itself rather than the quality of their language output. This is in line with Robinson's cognition hypothesis (2011), which posits that human attentional resources are limited and that as learners deplete these resources, the accuracy, fluency, or complexity of their language output may decline.

We also found that students in the AR-AFPA group performed better than those in the AR-TFPA group in terms of total writing scores, accuracy, and organization. This finding is supported by Liu et al. (2023), who argued that integrating AWE and FPA can produce more detailed diagnostic corrective feedback covering aspects such as grammatical accuracy, syntactic coherence, and punctuation norms. The immediate and direct corrective feedback provided by AWE helps students recognize and internalize correct language forms (Cheng & Zhang, 2024). In addition, the iterative submission process in the AR-AFPA approach promotes learner autonomy. According to skill acquisition theory, this autonomy enables learners to reinforce and automate accurate linguistic structures through repeated practice of language structure knowledge internalization, error modification, and correct knowledge consolidation (Cheng & Zhang, 2024). This process advances overall writing skills beyond linguistic accuracy alone. In terms of organization, our findings are consistent with those of Link et al. (2022) but different from those of Fu et al. (2022), who suggested that students may focus more on scoring and make micro-level revisions at the word and sentence levels when reviewing AWE feedback, while potentially neglecting higher-level revisions such as content organization and logical structure. These discrepancies may be attributed to different levels of learners' HOT. The AR-AFPA approach fosters HOT skills such as reflection and critical thinking through FPA activities (Filius et al., 2018), which in turn encourages learners to reflect critically on AWE feedback, thereby maximizing the benefits of detailed feedback and fostering a preference for global revisions (Liu et al., 2023). However, no significant differences were observed between the two groups in expression and creativity. Notably, the mean creativity score of the AR-AFPA group ($M = 76.89$) was slightly

lower than that of the AR-TFPA group ($M = 78.03$). One possible explanation is that AWE feedback in the AR-AFPA approach tended to follow a standardized format, reducing opportunities for interaction and meaning negotiation, which may have constrained learners' creative expression in writing (Li et al., 2025).

5.2 Effect of learning approaches on HOT

This study confirmed that the AR-TFPA approach was more effective than the TFPA approach in terms of improving students' HOT. However, significant differences were only observed in the problem solving dimension of the divergent HOT, with the AR-TFPA group outperforming the TFPA group. No significant differences were found in the other three dimensions: critical thinking, metacognition, and creativity. Flower and Hayes (1977) described writing as a complex problem-solving activity that requires constant decision making and challenge resolution. Previous studies have shown that AR technology, through its visual representation and interactive features, can reduce cognitive load and make thinking processes concrete. This allows students to visualize and reflect on their problem-solving strategies, which in turn helps reactivate their HOT abilities, especially their problem-solving skills, when faced with new challenges, resulting in more creative solutions (Dunleavy et al., 2009). Although the AR-TFPA group performed better in critical thinking, metacognition, and creativity, these differences were not statistically significant. This may be due to the fact that both groups used FPA strategies, which maintained high levels of these aspects in both groups. Previous studies have indicated that when receiving feedback, FPA involves convergent HOT processes such as reflective thinking, critical thinking, and self-regulation (Alemdag & Yildirim, 2022). Additionally, providing suggestions and engaging in argumentative dialogues may activate divergent HOT processes such as creativity and problem solving (Zhan et al., 2023). Thus, it can be inferred that the observed differences in the problem-solving abilities of the AR-TFPA group and the TFPA group can be attributed to the influence of AR technology.

However, no significant differences were found between the AR-AFPA and AR-TFPA group in overall HOT skills or in any specific dimensions of convergent and divergent HOT. A plausible explanation is that both groups utilized AR learning materials and adopted FPA strategies. Notably, the AR-AFPA group showed slightly higher scores in both convergent and divergent HOT, suggesting that integrating AWE into the FPA process may aid the development of students' HOT. The lack of significant differences between the two groups may be due to the short intervention duration in this study. Li (2023) supported this view, indicating that the longer the intervention duration, the more evident the positive effects of AWE, and emphasized the reinforcing role of interpersonal interaction on AWE in the FPA process. However, Li (2023) also cautioned against overgeneralizing this result due to the limited scope of existing research. Therefore, further research is needed to investigate the effects of extending the AWE intervention period and teacher feedback on the effectiveness of FPA strategies in fostering students' HOT.

5.3 Mediating roles of received feedback

Regarding the third research question, we found that the impact of the learning approach on students' writing performance and HOT improvements before and after FPA was partially mediated by the feedback received. For the effects of the TFPA (vs. AR-TFPA) approach on writing performance improvement, Bootstrap analysis revealed that the amount of negative affective feedback served as a suppressor. Specifically, students in the TFPA (vs. AR-TFPA) approach received less negative feedback. This may be because AR-based learning often involves higher-level writing skills or performance, leading to elevated expectations and standards among peers. When these expectations are not met, students are more likely to provide negative feedback. Conversely, the reduction in negative feedback in the TFPA (vs. AR-TFPA) approach may mitigate its negative impact on writing performance improvement (Yu et al., 2021). Therefore, the suppressive effect of negative feedback likely arises from lower expectation levels in the TFPA approach. However, within cognitive feedback, no feedback types acted as a mediator in this relationship.

In the case of the AR-AFPA (vs. AR-TFPA) approach, we identified clear mediating effects on writing performance improvement. Within affective feedback, the amount of neutral feedback was the sole mediator. This aligns with prior research indicating that the AR-AFPA (vs. AR-TFPA) approach tends to promote the provision of neutral feedback (Barrot, 2023), which has been shown to sustain long-term writing skills improvement (Link et al., 2022). In the AR-AFPA approach, AWE employs pre-designed algorithms and evaluation criteria to assess students' texts, thereby minimizing the subjectivity and personal biases that are often inherent in human evaluations. As a result, the feedback it provided is more neutral and objective. As stressed by Chen et al. (2024), an excessive provision of either positive or negative feedback tends to trigger students' resistance, reducing their willingness to adopt the feedback and ultimately leading to less overall improvement. In contrast, neutral feedback is more readily accepted by lowering students' defensiveness, allowing them to focus on refining their writing and achieving continuous improvement. Furthermore, we identified mediating effects within cognitive feedback, where the amount of suggestive and didactic feedback jointly explained the primary influence of the AR-AFPA (vs. AR-TFPA) approach on writing performance improvement. By uncovering these mechanisms, we extend the findings of Liu et al. (2023), who focused only on direct effects without exploring the underlying mechanism of how these feedback types influence these effects.

Regarding HOT improvement, no significant mediators were identified for the TFPA (vs. AR-TFPA) approach. This supports our findings in the previous section, suggesting that feedback under different approaches has limited explanatory power for HOT improvement. However, for the AR-AFPA (vs. AR-TFPA) approach, clear mediating effects were found. As predicted by Hattie and Timperley (2007), the amount of neutral feedback within affective feedback served as a mediator. Unlike simple praise or criticism, neutral feedback primarily comprises factual statements and constructive suggestions, requiring students to actively analyze, reflect, and develop improvement strategies. This process cultivates critical thinking, meta-cognitive skills, and creative problem-solving abilities, thereby promoting HOT

improvement. In other words, by fostering a supportive environment that encourages independent reasoning and meaningful self-reflection, neutral feedback facilitates deeper cognitive engagement. Additionally, suggestive feedback was the sole mediator within cognitive feedback, further explaining the influence of the AR-AFPA (vs. AR-TFPA) approach on HOT improvement.

6 Implications and conclusions

This study contributes to the existing research in two critical ways. First, it proposed a six-stage writing learning model that combined AR, FPA, and AWE to enhance the writing process, offering new insights into the integrated application of these interventions in writing instruction. On one hand, AR offers students more writing stimulation by visualizing abstract concepts, providing contextual linguistic scaffolding, and supporting personalized interaction. Consequently, AR materials facilitate the initiation of writing, enrich students' ideas, and guide the organization of their work. On the other hand, the integration of AWE and FPA affords students the opportunity to compare and synthesize subjective peer feedback with objective AWE feedback, followed by discussing and questioning the differences or consistencies between them. This interactive process, along with the cognitive conflicts it generates, stimulates students' reflection and learning, thereby promoting the development of their HOT. Second, the current study demonstrated the mediating role of feedback types in the relationships between learning approaches and writing performance/higher-order thinking, which enhances our further understanding of how feedback types work in writing. These practices provide crucial insights for further refinement and implementation of writing learning and evaluating activities.

Furthermore, our findings provide valuable practical implications for educators in writing instruction. First, AR is a useful tool for supporting writing by enriching learners' ideas for writing. Teachers can integrate AR materials in a blended approach to assist writing based on the specific writing topic. For example, the writing topics related to outdoor discovery activities are more suitable for using AR materials, whereas paper-based writing scaffolding that provides rhetoric and syntax guidance may be better for descriptive essays (Wang, 2017). Second, the integration of automated feedback technology into FPA not only improves feedback efficiency, but also encourages students to reflect on peer feedback and writing revision, preventing them from merely skimming peer feedback without deep reflection. In the future, teachers should explore other AI tools to complement human writing assessment rather than relying solely on traditional evaluation methods. Moreover, our findings revealed that both neutral feedback and suggestive feedback mediate the relationships between learning approaches and improvements in writing performance or HOT. Therefore, to enhance students' writing skills and HOT, teachers should encourage the provision of feedback that includes factual statements and constructive suggestions, while minimizing feedback that includes excessive personal emotions, whether positive or negative.

In conclusion, this study examined the effects of different FPA-based learning approaches on students' writing learning. Specifically, integrating AR technology

into FPA activities significantly improved students' writing performance and HOT. While in an AR environment, students benefitted more from FPA that included AWE than from FPA incorporating teacher assessment. Furthermore, the influence of the AR-AFPA (vs. AR-TFPA) learning approach on students' writing performance and HOT improvements was partially mediated by the amount of neutral and suggestive feedback received in the affective and cognitive dimensions. In contrast, the mediation effects in the TFPA (vs. AR-TFPA) approach was negligible.

7 Limitations and recommendations for future research

Although this study provides a comprehensive analysis of the effects of different FPA-based learning approaches on writing instruction, some limitations must be acknowledged. First, the main limitation is the small number of participants and the duration of the experiment (5 weeks), which may not have been sufficient for the long-term development of writing and HOT skills. Although significant inter-group differences were observed within this period, these results might shift with extended observations. Second, this study investigated the effectiveness of the AR-based formative peer assessment learning mode and automated writing evaluation in writing instruction, focusing on a sample consisting solely of Chinese pupils. As a result, its ability to verify the universal applicability of these interventions in writing instruction is limited. Additionally, while higher-order thinking is known to significantly influence writing performance (Wu et al., 2024), this study did not delve into the complex interactions between these factors. Furthermore, the focus was solely on the mediating effects of feedback types, without exploring the detailed characteristics of feedback such as length, quantity, and quality. Finally, the integration of information and communication technologies (ICT) such as AR and AWE into diverse educational contexts may face implementation challenges, including gaps in infrastructure readiness, accessibility issues, and the need for adequate teacher training.

To address the limitations in the study, it would be expedient to: 1) Include in future studies an assessment of the long-term effects of using AR, FPA, and AWE on academic achievement and higher-order thinking, which will allow us to understand whether positive changes in writing skills remain stable over time; 2) Consider the cultural and geographical specificity of the sample, which will further validate the generalizability of these intervention strategies across diverse writing learning contexts; 3) Investigate the relationships between different HOT skills and specific writing achievements, which could offer clearer insights into how these skills influence writing performance; 4) Consider other feedback features and cognitive factors, such as motivation level, learning style, and individual information processing characteristics, that may affect the results of using these interventions and deepen our understanding of how they influence writing instruction; 5) Explore the use of cloud computing to make AR and AWE tools more accessible via online platforms to reduce hardware costs and technical barriers, thus addressing the challenges associated with applying ICT in education; and 6) Organize regular teacher training sessions and workshops could enhance educators' understanding of the practical applications and operational aspects of AR and AWE.

Appendix

Table 6 Rubric for Chinese writing performance

Dimension	Index	85–95	75–85	65–75	65–55
Accuracy	The precise grasp of content, accurate word usage, standard punctuation, and the absence of obvious grammatical errors	The content aligns with the topic, central theme is prominent, and narrative flows smoothly. No typos, correct punctuation enhances emotional expression	The content stays on topic and aligns with the theme. Despite 1–2 typographical errors, punctuation is used effectively for expression	The content maintains a generally clear theme, with 3–5 typos and standard punctuation usage	The content exhibits a biased understanding of the topic, lacks clarity in central ideas, and diverges from the intended writing theme. Numerous typos hinder reader comprehension, compounded by improper punctuation usage
	Organization	The structure of the written composition is complete, details are appropriate, paragraphs are reasonable, and basic elements of writing can be expressed	The structure of the written composition is well organized, coherent, and thoughtful with a logical and analytical progression of ideas	The structure of the written composition is slightly loose. There is some inconsistency but the overall organization is coherent	The structure of the written composition is not clear with incorrect paragraphing
Expression	The ability to utilize rhetorical devices and sensory verbs to convey emotions and details in writing	Precisely uses various analogies, similes, or metaphors to describe the features of the subject (over 5 instances)	Appropriately uses several analogies, similes, or metaphors to describe the features of the subject. (3–5 instances)	Attempts to use several analogies, similes, or metaphors to describe the features of the subject. (1–2 instances)	Fails to use or incorrectly uses analogies, similes, or metaphors to describe the features of the subject

Table 6 (continued)

Dimension	Index	85–95	75–85	65–75	65–55
Creativity	The theme, narrative perspective, and evidence description are unique and novel	The writing features creative and original ideas, embodying a uniquely profound conceptualization with a distinctive narrative perspective. Several instances (3 or more) of evidence depiction use expressive techniques like association, imagination, analogy, and citation	The writing features moderate creativity and originality. While the conceptualization is somewhat profound, it lacks distinctiveness, and the narrative perspective is conventional. Several instances (2–3) of evidence presentation use expressive techniques such as association, imagination, analogy, and citation	The writing attempts to convey creative and original ideas. However, the conceptualization lacks depth and uniqueness, and the narrative perspective is conventional. Only a few instances (1–2) of evidence presentation use expressive techniques such as association, analogy, and citation	The writing fails to produce creative and original ideas. The conceptualization of the writing is unclear, the narrative perspective is confused, and there is a lack of evidence employing expressive techniques such as association, imagination, analogy, and citation

Table 7 The higher-order thinking (HOT) questionnaire

Dimension	Item
Divergent HOT	
Problem solving	1. I believe that when writing, it is important to clarify the logical flow of thoughts and express emotions
	2. In addition to knowing the steps to writing, it is important to understand the reasons associated with the steps
	3. When I encounter a problem that needs to be solved while writing, I first think of ways to solve the problem
Creative thinking	4. In this class, I generate many new ideas for writing
	5. In this class, I create different solutions for a problem when writing
	6. In this class, I produce ideas that are likely to be useful
Convergent HOT	
Metacognition	7. When I take writing courses, I ask myself periodically whether I am meeting my goal
	8. I regularly review my knowledge of writing
	9. When I complete my writing courses, I ask myself how well I accomplished my goals
	10. When I complete my writing courses, I ask myself if I have learned as much as I could
Critical thinking	11. In writing classes, I have learnt more about how to approach complex issues in a variety of ways
	12. In writing classes, I have developed a more focused and systematic way of thinking
	13. In writing classes, I have improved my ability to judge the value of new information or evidence presented to me
	14. In writing classes, I have learnt more about how to analyze the key issues in my writing
	15. In writing classes, I have improved my ability to give sound reasons for my beliefs, opinions and ideas
	16. In writing classes, I have developed a more open-minded approach to interpreting, analyzing and judging alternative points of view

Table 8 Validity and reliability results of the higher-order thinking questionnaire

Dimensions	Items	Loadings	Cronbach's alpha (α)	McDon- ald's Omega (ω)	Composite reliability (CR)	Average variance extracted (AVE)
Problem solving	1	0.602	0.802	0.805	0.580	0.802
	2	0.800				
	3	0.866				
Creative thinking	1	0.808	0.862	0.861	0.674	0.862
	2	0.875				
	3	0.811				
Metacognition	1	0.721	0.878	0.881	0.650	0.878
	2	0.800				
	3	0.736				
	4	0.686				
Critical thinking	1	0.722	0.856	0.858	0.503	0.856
	2	0.757				
	3	0.721				
	4	0.639				
	5	0.678				
	6	0.649				

Table 9 Coding scheme for the feedback messages

Dimension	Category	Definition	Example
Affective (A)	(A1) Positive	Positive feelings about, support for, or praise for the assessed writing	“Well written!”
	(A2) Negative	Negative or unfavorable feelings about or criticism for the assessed writing	“The writing is very incoherent and lacks accuracy.”
	(A3) Neutral	There were no critical, affirming, or praising feelings about the assessed writing	“Some commas can be changed to periods.”
Cognitive (C)	(C1) Corrective feedback	Corrective feedback is offered when assessors point out incorrect information in the assessed writing or correct it directly. This kind of feedback can effectively reduce errors or inaccuracies in students’ compositions	“The underlined sentence in the first paragraph can be changed to ‘I find the climate here is hot and humid.’”
	(C2) Reinforcing feedback	Reinforcement feedback is provided when students demonstrate correct or appropriate writing. It conveys positive feelings or recognition of assessed compositions. This kind of feedback sometimes occurs when students are encouraged without explicitly knowing the reasons behind it	“Although the number of words in your composition is a bit short, it is still very good overall.”
	(C3) Didactic feedback	Didactic feedback is given when the assessor offers a lengthy explanation to assist the assessee who has made errors or has provided inadequate information. This kind of feedback usually involves lengthy explanations with a lecture tone, aiming to guide students in the right direction	“From a grammatical perspective, ‘necessary’ functions as an adjective, often combined with other words in a sentence to attribute qualities; whereas ‘must’ serves solely as an auxiliary verb, typically conveying obligation or necessity.”
	(C4) Suggestive feedback	Suggestive feedback is provided if assessed writing is incomplete rather than incorrect. The assessor may alert the student that there is a problem without specifically identifying the issue. Such feedback can take the form of hints to guide the student’s thinking. This kind of feedback is also considered a form of scaffolding	“It is recommended that the characteristics of the tiger be described in more detail.”
	(C5) Irrelevant feedback	Feedback that is not related to writing content	“I often use a book as a makeshift writing pad.”

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Data availability The data and materials are available upon request to the corresponding author.

Code availability Not applicable.

Declarations

Ethics approval The ethical requirements for research in this selected university were followed.

Consent to participate The participants all agreed to take part in this study.

Consent for publication The publication of this study has been approved by all authors.

Conflicts of interest There is no potential conflict of interest in this study.

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